Enhancing Performance Constraints of Nonlinear Networked Systems Under Intermittent Communications and DoS Attacks

Xinyu Xu, Xinjun Wang, *Member, IEEE*, Yingying Liu, *Member, IEEE*, Liang Zhang, and Ben Niu

***Abstract*—This article mainly studies the problem of intermittent output feedback tracking algorithm for a class of nonlinear networked systems with monotonic tube boundary functions (MTBFs) constraints under denial-of-service (DoS) attacks. Intermittent malicious DoS attacks result in the output signal and states of the system being unavailable, increasing the complexity of controller design. To handle this issue, we create a novel composite observer based on an attack compensator, which utilizes the intermittent output signal to reconstruct the state variables of the system. Then, the dynamic filtering technology is introduced to solve the problem of the virtual controller non-differentiability caused by intermittent output feedback in the backstepping design process. On this basis, a replacement scheme is proposed: firstly, a continuous control strategy is designed based on conventional output feedback, and then the states in the continuous control scheme were replaced with triggered states to construct an event-triggered control scheme. Additionally, a set of MTBFs is developed to avoid excessive overshoot or jitter of system errors in transient performance. The results indicate that the proposed event-triggered safety control strategy can ensure the boundedness of all signals in the closed-loop system, while converging the tracking error to a prescribed range near the origin. Finally, the simulation result from a single-link robot confirmed the effectiveness of the proposed strategy.**

***Index Terms*—Backstepping, nonlinear systems, monotone tube boundary functions, denial-of-service attacks, event-triggered.**

I. Introduction

IN RECENT years, with the widespread application of network technology in various fields, the networked control systems (NCS) have received increasing attention due to their flexibility, intelligence and scalability [1], [2], [3], [4], [5], [6], [7]. The NCS is a system that connects sensors, controllers, and actuators through communication networks. These components exchange information via wired or wireless links to enable resource sharing. However, due to the NCS relying on open communication networks, it is vulnerable to cyber attacks such as denial-of-service (DoS) attacks [8], [9], [10], [11], deception attacks [12], [13], [14], [15] and replay attacks [16], [17], [18]. Among them, DoS attacks occupy the bandwidth, processing power, or other resources of the target system, causing the system to be unable to provide services normally, thereby interrupting or stopping services, and ultimately damaging the stability and performance of the system. Therefore, how to effectively respond to DoS attacks has become an important research topic in the field of the NCS. In [19], an adaptive control scheme for uncertain nonlinear systems was proposed by analyzing the impact of DoS attack duration, frequency, and rest time on system stability. In [20], an adaptive fuzzy decentralized controller based on output feedback was designed for nonlinear large-scale network physical systems that are susceptible to DoS attacks. Although the above studies have made significant progress in addressing DoS attacks, they often do not fully consider communication efficiency and resource constraints in practical NCS. Therefore, achieving efficient control and communication under limited resources while ensuring system stability and security remains a critical issue in current research.

The event-triggered control (ETC) strategy effectively addresses the aforementioned issues. Compared with traditional time-driven control schemes, the ETC only performs data transmission and control updates when the system state meets specific triggering conditions, significantly reducing the consumption of communication resources [21], [22], [23], [24]. This strategy not only reduces network bandwidth pressure but also effectively reduces the risk of exposure to network attacks due to frequent communication. But this can also lead to a certain amount of data loss, thereby reducing the performance of the system. Therefore, many researchers have conducted extensive research on the ETC. In [25], a fuzzy-based ETC scheme is developed for non-strict feedback nonlinear systems with fixed-time convergence. In [26], an event-triggered adaptive fixed-time control scheme is proposed for uncertain nonlinear systems with unknown virtual control coefficients. However, most existing studies rely on ETC schemes with fully available state information,

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whereas designing ETC schemes based solely on output signals is more challenging and of practical significance. In [27], an adaptive event-triggered finite-time control scheme is proposed for state-unpredictable switched nonlinear systems. In [28], a state observer and adaptive law are constructed using only intermittent outputs for nonlinear systems, and an adaptive neural network based ETC scheme is proposed. It is worth noting that existing ETC strategies are mainly used for controller updates, while sensor transmissions remain continuous, leading to inefficient use of communication resources. Therefore, extending ETC strategies to sensor-to-controller communication is of great practical significance.

In addition, the improvement of system control performance is crucial for the stability and response speed of the system. Therefore, many researchers have devoted themselves to improving transient and steady-state performance and have achieved abundant research results [29], [30], [31], [32]. Among them, the most typical and common are the funnel control (FC) and the prescribed performance control (PPC). The FC method constrains the tracking error of the system by designing a dynamic funnel boundary, ensuring that the error always remains within the predefined bounds. The PPC method ensures that the transient and steady-state performance of the system meets design requirements through prescribed performance functions. Although FC and PPC methods perform well in improving system control performance, they may exhibit overshoot and jitter behavior during transient phases. To address this issue, the monotonic tube boundary function (MTBF) was proposed in [33]. Compared with traditional FC and PPC, the monotonic tube control (MTC) can more smoothly constrain system errors, effectively reduce overshoot and jitter phenomena in the transient phase, and further improve the robustness and stability of control. However, in [33], only the use of MTBFs in a secure environment was studied, without considering the situation where the system is subjected to cyber attacks. Under DoS attacks, communication interruptions and data loss prevent MTC from maintaining prescribed error constraints, thereby posing challenges to performance analysis. Moreover, the simultaneous use of MTBFs and the event-triggered mechanism (ETM) further increases the complexity of controller design and stability analysis. Therefore, it remains a challenging task to design a resilient event-triggered tracking control strategy based on MTBFs for nonlinear systems subjected to cyber attacks.

Following the above discussion, this paper explores the tracking problem of a class of nonlinear networked systems under DoS attacks, using only intermittent output signals and constrained by MTBFs. The main contributions are summarized below.

1) Unlike [34] and [35], the output signal after triggering in this paper can only be intermittently received by the observer due to malicious DoS attacks, rendering traditional observer design ineffective. To address this issue, a novel composite observer based on an attack compensator is proposed by utilizing the triggered output signal. This observer not only reconstructs the system state variables but also handles unknown external disturbances, effectively compensating for the adverse effects of event-triggering and DoS attacks on the system.
2) In contrast to the FC methods in [29], [30], [36] and the PPC methods in [31], [32], [37], this paper proposes an MTC method based on hyperbolic cosecant functions. The proposed method guarantees a preassigned convergence time while effectively suppressing overshoot and jitter in the transient tracking process, thereby improving the system's robustness and stability.
3) A first-order filter is introduced in this paper to effectively avoid the explosion of complexity caused by the recursive differentiation of virtual controllers. Different from [38] and [39], an ETM based on the output derivative rather than the output signal is designed for the filter. Moreover, by exploiting the dynamic characteristics of the filter, the boundedness of the virtual controller triggering error is established, thereby overcoming the difficulty in handling the triggering error under the constraints of MTBFs.

II. Preliminaries and System Description

A. Problem Formulation

Consider the dynamic model of nonlinear networked systems as follows:

$$\begin{cases} \dot{x}_i = x_{i+1} + f_i(y) + d_i(t), i = 1, \cdots, n-1, \\ \dot{x}_n = u + f_n(y) + d_n(t), \\ y = x_1, \end{cases} \tag{1}$$

where $x_i (i = 1, \cdots, n)$ indicate the system state, $y \in \mathbb{R}$ represents the system output, $u \in \mathbb{R}$ represents the system input, $f_i(y)(i = 1, \cdots, n)$ represent the known smooth nonlinear function, $d_i(t)(i = 1, \cdots, n)$ represent the external disturbance, which is an unknown bounded continuous function.

Assumption 1: y_d is the desired trajectory of the system and satisfies n-order differentiability.

Assumption 2: The nonlinear function $f_i(y)(i = 1, \cdots, n)$ satisfies the local Lipschitz continuity condition, as shown below

$$|f_i(y_1) - f_i(y_2)| \leq l_i |y_1 - y_2|, \tag{2}$$

where l_i is a positive constant.

Assumption 3: The continuous time-varying disturbance $d_i(t)$ is bounded, and there exists a positive constant d_M such that $|\dot{d}_i(t)| \leq d_M$.

B. Event-Triggered Mechanism

In order to save communication resources, the event-triggered mechanism (ETM) is designed as follows:

$$\begin{aligned} &\bar{y}(t) = y(t_s), \quad t \in (t_s, t_{s+1}), \\ &t_{s+1} = \inf\{t > t_s, |y(t) - \bar{y}(t)| > \mathfrak{m}_1\}, \end{aligned} \tag{3}$$

where $\mathfrak{m}_1$ is the designed triggering threshold that satisfies $\mathfrak{m}_1 > 0$, $t_s (s = 0, 1, 2, \cdots)$ represents the sth event-triggered instant of the output signal, t_0 is the first triggering moment that satisfies condition (3).

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C. Intermittent DoS Attack

With the rapid development of the internet, cyber attacks have become increasingly complex. This article mainly studies intermittent DoS attacks by malicious adversaries on communication channels. Assuming T_j^a and T_j^b respectively represent the start and end times of the jth DoS attack, the interval between the jth DoS attacks can be obtained as $[T_j^a, T_j^b]$. The total interval of DoS attacks within $[\epsilon, t]$ is expressed as

$$\beth_a(\epsilon, t) = \left\{ \bigcup_{j \in \mathcal{N}} [T_j^a, T_j^b) \right\} \bigcap [\epsilon, t], \tag{4}$$

where $\mathcal{N}$ is a positive integer, $\epsilon > 0$ and $t > 0$ with $t > \epsilon$.

Then, the interval attack sleep time can be defined as

$$\beth_b(\epsilon, t) = [\epsilon, t] \backslash \beth_a(\epsilon, t). \tag{5}$$

Assumption 4 (DoS Duration [40]): There exists a constant $\mathcal{T}_1 \geq 1$, which makes the following equation hold

$$|\beth_a(\epsilon, t)| \leq \frac{t - \epsilon}{\mathcal{T}_1}.$$

Assumption 5 (DoS Frequency [40]): The existence of constants $\theta > 0$ and $\epsilon_D > 0$, which makes the following equation hold

$$\sigma(\epsilon, t) \leq \theta + \frac{t - \epsilon}{\epsilon_D},$$

where $\sigma(\epsilon, t)$ is the frequency of DoS attacks occurring.

D. Monotone Tube Boundary Functions

Definition 1: A pair of functions $\bar{\eta}$ and $\underline{\eta}$, which are continuous and exist continuous first derivatives, are called MTBFs if the following conditions are satisfied.

(i) $\bar{\eta}$ and $\underline{\eta}$ are monotonically increasing or decreasing functions, and for $T > 0$, $\bar{\eta}$ and $\underline{\eta}$ strictly monotonically increasing or decreasing on $[0, T]$;

(ii) $\bar{\eta}$ and $\underline{\eta}$ are bounded, and there exist constants $\bar{\eta}_\infty$ and $\underline{\eta}_\infty$ such that $\lim_{t \to \infty} \bar{\eta} = \bar{\eta}_\infty$ and $\lim_{t \to \infty} \underline{\eta} = \underline{\eta}_\infty$;

(iii) For all $t > 0$, $\bar{\eta}_\infty > \underline{\eta}_\infty$.

Remark 1: In order to better demonstrate the advantages of MTBFs, this paper compares it with traditional PPC methods, as shown in Fig. 1, where $z_1(0)$ represents the initial tracking error. Although both methods can keep the tracking error within the predetermined boundaries $(-p, p)$ and $(\bar{\eta}, \underline{\eta})$, the traditional PPC method cannot accurately regulate the convergence time and transient overshoot due to its "two-sided" exponentially decaying boundaries. In contrast, the MTBFs can better control the convergence time of errors, reduce overshoot in transient response, and provide more accurate system regulation by introducing a "one-sided" hyperbolic cosecant function, overcoming the shortcomings of traditional PPC methods.

Through the above analysis, we can know that when the initial value $z_1(0)$ of the system tracking error satisfies $\bar{\eta}(0) < z_1(0) < \underline{\eta}(0)$, MTC can ensure that $z_1(t)$ satisfies

$$\bar{\eta}(t) < z_1(t) < \underline{\eta}(t), t \geq 0 \tag{6}$$

where

$$z_1 = \begin{cases} \hat{x}_1 - y_d, & t \in [T_j^a, T_j^b) \\ x_1 - y_d, & t \in [T_j^b, T_{j+1}^a) \end{cases}$$

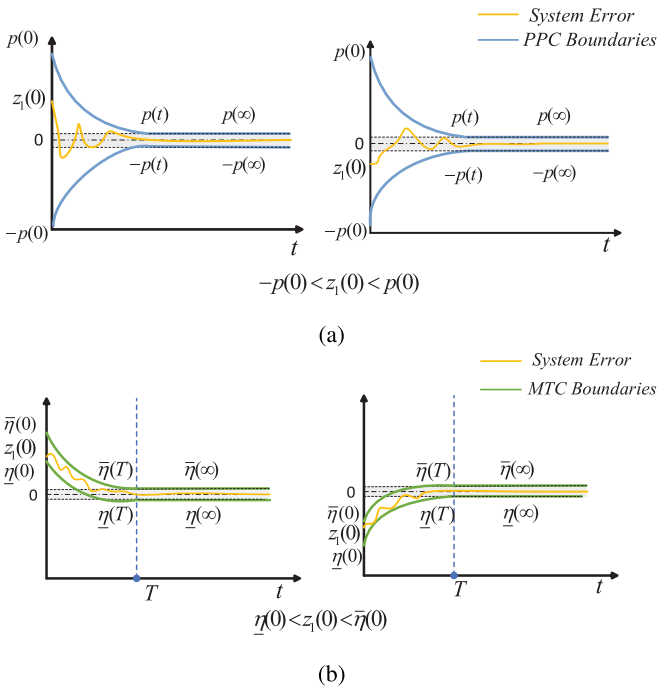


Fig. 1. Depiction of the traditional PPC method and the proposed MTC method in (a) and (b).

In order to further design the control scheme, a pair of feasible boundary functions were designed by exploiting the initial value and time derivative of the hyperbolic cosecant function csch, as shown below:

$$\mathrm{col}(\bar{\eta}, \underline{\eta}) = \mathrm{sign}(z_1(0))(\rho - \rho_\infty)\mathrm{col}(1, 1) + \Re\rho, \tag{7}$$

with $\Re = \mathrm{diag}(\bar{\gamma}, -\underline{\gamma})$ and

$$\rho = \begin{cases} \mathrm{csch}\left(\rho_0 + \dfrac{\imath t}{T - t}\right) + \rho_\infty, & 0 \leq t < T, \\ \rho_\infty, & t \geq T. \end{cases} \tag{8}$$

where $\imath \in R_{>0}$ represents the convergence rate of the error z_1, ρ_0 represents the initial value of the preselected boundary, and $\bar{\gamma}, -\underline{\gamma}, \rho_\infty \in R_{>0}$. Evidently, starting from instance T, the value of MTBFs will gradually approach the preselected values $\bar{\eta}_\infty = \bar{\gamma}\rho_\infty$ and $\underline{\eta}_\infty = -\underline{\gamma}\rho_\infty$.

Remark 2: By differentiating (8), we can obtain

$$\dot{\rho} = \begin{cases} -\dfrac{\cosh\left(\rho_0 + \frac{\imath t}{T-t}\right)\imath T}{\sinh^2\left(\rho_0 + \frac{\imath t}{T-t}\right)(T - t)^2}, & 0 \leq t < T, \\ 0, & t \geq T. \end{cases} \tag{9}$$

Based on (9), we can infer that $\dot{\rho} < 0$, $\forall t > 0$, that is, $\lim_{m \to +\infty} \rho(m) < \rho(t) < \rho(0)$, which guarantees the boundedness of ρ. Furthermore, we infer that $\lim_{m \to T_k^-} \dot{\rho}(m) = \lim_{m \to T^+} \dot{\rho}(m) = \dot{\rho}(T) = 0$, ensuring the continuity of the first derivatives of functions ρ, $\bar{\eta}$ and $\underline{\eta}$.

III. Main Results

A. Compensator-Based Composite Observer Design

The system affected by DoS attacks is illustrated in Fig. 2. When the communication link between the sensor and the

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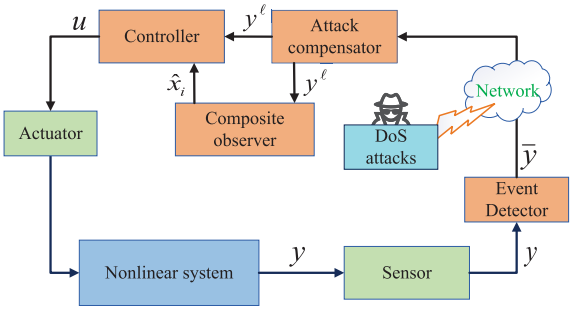


Fig. 2. Control diagram of nonlinear system with DoS attacks.

controller is subjected to DoS attacks, the information transmission over this link may be interrupted. Moreover, the ETM transmits sensor data only when the triggering condition is satisfied. To mitigate the adverse effects caused by intermittent information transmission, a novel composite observer based on an attack compensator is designed to reconstruct the state variables.

Let y^ℓ represent the system's output signal under DoS attacks, the attack compensator can be designed as

$$y^\ell = \begin{cases} \bar{y}(T_j^{a-}), & t \in [T_j^a, T_j^b) \\ \bar{y}, & t \in [T_j^b, T_{j+1}^a) \end{cases} \tag{10}$$

where T_j^{a-} represents the left limit of T_j^a.

Remark 3: It should be noted that when the occurrence instant of a DoS attack coincides with an event-triggering instant, the communication link is blocked, and the newly triggered data cannot be successfully transmitted to the observer. Therefore, the observer performs state estimation solely based on signal $y(T_j^{a-})$. The newly triggered data can be used for observer state updates only when the communication link is restored.

Considering the unmeasurable system state under DoS attacks, the composite observer based on attack compensator is designed as

$$\begin{cases} \dot{\hat{x}}_i = \hat{x}_{i+1} + f_i^\varrho + m_i(y^\ell - \hat{x}_1) + \hat{d}_i, \\ \dot{\hat{x}}_n = u + f_n^\varrho + m_n(y^\ell - \hat{x}_1) + \hat{d}_n, \\ \hat{y} = \hat{x}_1, i = 1, \cdots, n-1 \end{cases} \tag{11}$$

where m_i denote the design parameters, $\hat{x}_i$ represent the estimate of x_i, $\hat{d}_i$ represent the estimate of d_i, $\tilde{d}_i = d_i - \hat{d}_i i = 1, \cdots, n$. Using ϱ as the indicator factor to indicate whether the system has suffered from DoS attacks, that is, if $\varrho = a$, then system (1) is under attack, if $\varrho = b$, then system (1) is safe, and f_i^ϱ is a bounded known function

$$f_i^\varrho = \begin{cases} f_i(\hat{y}), & \varrho = a \\ f_i(\bar{y}), & \varrho = b \end{cases} \tag{12}$$

The estimate error is expressed as $e_i = x_i - \hat{x}_i, i = 1, \cdots, n$. Based on (1) and (11), we have

$$\dot{e} = \begin{cases} \mathcal{A}_a e + \triangle\mathcal{F}_a - \mathcal{M}_a(\bar{y}(T_j^{a-}) - \hat{x}_1) + \tilde{\mathcal{D}}, \\ \quad t \in [T_j^a, T_j^b), \\ \mathcal{A}_b e + \triangle\mathcal{F}_b - \mathcal{M}_b(\bar{y} - x_1) + \tilde{\mathcal{D}}, \\ \quad t \in [T_j^b, T_{j+1}^a), \end{cases} \tag{13}$$

where $e = [e_1, \cdots, e_n]^T$, $\tilde{\mathcal{D}} = [\tilde{d}_1, \cdots, \tilde{d}_n]^T$, $\mathcal{M}_a = [m_{1a}, \cdots, m_{na}]^T, \mathcal{M}_b = [m_{1b}, \cdots, m_{nb}]^T, \triangle\mathcal{F}_a = [f_1(y) - f_1(\hat{y}), \cdots, f_n(y) - f_n(\hat{y})]^T$, $\triangle\mathcal{F}_b = [f_1(x_1) - f_1(\bar{x}_1), \cdots, f_n(y) - f_n(\bar{y})]^T$ and $\mathcal{A}_b = \mathcal{A}_a - \mathcal{M}_b\mathcal{C}$ with $\mathcal{C} = [1, 0, \cdots, 0]^T$, $A_a = \begin{bmatrix} 0 & I_{n-1} \\ 0 & 0 \end{bmatrix}$.

In addition, to handle the unknown disturbance $d_i(t)$, an intermediate variable is designed as follows

$$\Phi_i = d_i - \omega_i x_i, i = 1, \cdots, n \tag{14}$$

where the design parameter $\omega_i > 0$ is a constant, during DoS attacks, Φ_i is represented as Φ_{ia} and ω_i as ω_{ia}; while in a secure situation, Φ_i is represented as Φ_{ib} and ω_i as ω_{ib}. By taking the derivative of Φ_i, we can obtain

$$\begin{aligned} \dot{\Phi}_i &= \dot{d}_i - \omega_i[x_{i+1} + f_i(y) + \Phi_i + \omega_i x_i], \\ \dot{\Phi}_n &= \dot{d}_n - \omega_n[u + f_n(y) + \Phi_n + \omega_n x_n]. \end{aligned} \tag{15}$$

The derivative of the estimation of auxiliary variable $\hat{\Phi}_i$ can be designed

$$\begin{aligned} \dot{\hat{\Phi}}_i &= -\omega_i[\hat{x}_{i+1} + f_i^\varrho + \hat{\Phi}_i + \omega_i \hat{x}_i], \\ \dot{\hat{\Phi}}_n &= -\omega_n[u + f_n^\varrho + \hat{\Phi}_n + \omega_n \hat{x}_n]. \end{aligned} \tag{16}$$

Based on the above analysis, the estimation of external disturbance $\hat{d}_i$ can be expressed as $\hat{d}_i = \hat{\Phi}_i + \omega_i \hat{x}_i$. Let $\tilde{\Phi}_i = \Phi_i - \hat{\Phi}_i$, and using (15) and (16), we can obtain

$$\begin{aligned} \dot{\tilde{\Phi}}_i &= \dot{d}_i - \omega_i[e_{i+1} + (f_i(y) - f_i^\varrho) + \tilde{\Phi}_i + \omega_i e_i] \\ \dot{\tilde{\Phi}}_n &= \dot{d}_n - \omega_n[u + (f_n(y) - f_n^\varrho) + \tilde{\Phi}_n + \omega_n e_n]. \end{aligned} \tag{17}$$

For $t \in [T_j^a, T_j^b)$, we define $e_a = \bar{y}(T_j^{a-}) - \hat{x}_1$, then, the Lyapunov function V_0^a can be selected as

$$V_0^a = e^T P_a e + \frac{1}{2}\sum_{i=1}^{n} \tilde{\Phi}_{ia}^2, \tag{18}$$

where $P_a = P_a^T > 0$.

Based on (10), (13), (17) and (18), we can get

$$\begin{aligned} \dot{V}_0^a = {} & e^T(A_a^T P_a + P_a A_a)e + 2e^T P_a(\triangle\mathcal{F}_a - \mathcal{M}_a e_a + \tilde{\mathcal{D}}) \\ & + \sum_{i=1}^{n-1} \tilde{\Phi}_{ia}[\dot{d}_i - \omega_{ia}(e_{i+1} + (f_i(y) - f_i(\hat{y})) + \tilde{\Phi}_{ia} \\ & + \omega_{ia} e_i)] + \tilde{\Phi}_{na}[\dot{d}_n - \omega_{na}(\tilde{\Phi}_{na} + (f_n(y) - f_n(\hat{y})) \\ & + \omega_{na} e_n)]. \end{aligned} \tag{19}$$

According to Young's inequality and (19), one gets

$$\begin{aligned} \dot{V}_0^a \leq {} & e^T \Bigg(A_a^T P_a + P_a A_a + \frac{1}{k} + k\|P_a \mathcal{L}_a\|^2 + \frac{1}{c} + c\|P_a \mathcal{M}_a\|^2 \\ & + 2P_a^T P_a + \frac{3}{2}\sum_{i=1}^{n} \omega_{ia}^2 + \frac{1}{2}\sum_{i=1}^{n} \omega_{ia}^2 l_{ia}^2 + \frac{1}{2}\sum_{i=1}^{n} \omega_{ia}^4 \Bigg) e \\ & - \sum_{i=1}^{n} (\omega_{ia} - 3)\tilde{\Phi}_{ia}^2 + \mathfrak{V}_a. \end{aligned} \tag{20}$$

where $\mathfrak{V}_a = \frac{n}{2} d_M$ and $\mathcal{L}_a = [l_{1a}, \cdots, l_{na}]$.

For $t \in [T_j^b, T_{j+1}^a)$, we define $e_b = \bar{y} - \hat{x}_1$, then, the Lyapunov function V_0^b can be selected as

$$V_0^b = e^T P_b e + \frac{1}{2}\sum_{i=1}^{n} \tilde{\Phi}_{ib}^2. \tag{21}$$

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where $P_b = P_b^T > 0$.

Based on (10), (13), (17) and (21), we can get

$$\begin{aligned}\dot{V}_0^b = {} & e^T(A_b^T P_b + P_b A_b)e + 2e^T P_b(\triangle \mathcal{F}_b + \mathcal{M}_b(\bar{y} - y) + \tilde{\mathcal{D}}) \\ & + \sum_{i=1}^{n-1} \tilde{\Phi}_{ib}[\dot{d}_i - \omega_{ib}(\tilde{x}_{i+1} + (f_i(y) - f_i(\bar{y})) + \tilde{\Phi}_{ib} \\ & + \omega_{ib} e_i)] + \tilde{\Phi}_{nb}[\dot{d}_n - \omega_{nb}(\tilde{\Phi}_{na} + (f_n(y) - f_n(\bar{y})) \\ & + \omega_{nb} e_n)]. \end{aligned} \tag{22}$$

According to Young's inequality and (22), we can obtain

$$\begin{aligned}\dot{V}_0^b \le {} & e^T \Bigg(A_b^T P_b + P_b A_b + P_b \mathcal{L}_b \mathcal{L}_b^T P_b + P_b \mathcal{M}_b \mathcal{M}_b^T P_b \\ & + 2P_b^T P_b + \frac{3}{2}\sum_{i=1}^{n} \omega_{ib}^2 + \frac{1}{2}\sum_{i=1}^{n} \omega_{ib}^4 \Bigg) e \\ & - \sum_{i=1}^{n} (\omega_{ib} - 3)\tilde{\Phi}_{ib}^2 + \mathfrak{V}_b, \end{aligned} \tag{23}$$

where $\mathfrak{V}_b = 2\mathfrak{m}_1 + \frac{1}{2}\sum_{i=1}^{n} \omega_{ib}^2 l_{ib}^2 \mathfrak{m}_1 + \frac{n}{2} d_M$ and $\mathcal{L}_b = [l_{1b}, \cdots, l_{nb}]$.

B. Resilient Event-Triggered Controller Design

In this section, we propose a secure tracking control strategy with an ETM based on the backstepping method for system (1). Based on the observer (11) and the first-order filter, we can obtain the following coordinate transformation:

$$z_1 = \chi - y_d, z_k = \hat{x}_k - \theta_{kf}, \nu_k = \theta_{kf} - \alpha_{k-1}, \tag{24}$$

with

$$\chi = \begin{cases} \hat{x}_1, & t \in [T_j^a, T_j^b) \\ x_1, & t \in [T_j^b, T_{j+1}^a) \end{cases} \tag{25}$$

where $k = 2, \cdots, n$, $z_k (k = 1, \cdots, n)$ is error variable, α_{k-1} represents the virtual control variable, θ_{kf} represents the output of the first-order filter, and ν_k refers to the output error of the first-order filter, we can get

$$\tau_k \dot{\theta}_{kf} + \theta_{kf} = \alpha_{k-1}, \theta_{kf}(0) = \alpha_{k-1}(0), \tag{26}$$

where $\tau_{i,k} > 0$ is a constant.

To obtain the main results, the specific design process is as follows.

***Case 1*:** When $t \in [T_j^b, T_{j+1}^a)$, the system did not suffer from DoS attacks. In order to simplify the design of the tracking control strategy, we first designed a continuous control method without ETM, and the tracking error (24) becames

$$z_1^b = x_1 - y_d, z_k^b = \hat{x}_k - \theta_{kf}^b, \nu_k^b = \theta_{kf}^b - \alpha_{k-1}^b, \tag{27}$$

where $k = 2, \cdots, n$, $z_k^b (k = 1, \cdots, n)$, α_{k-1}^b, θ_{kf}^b and ν_k^b represent the errors of the system, the virtual controller, the output of the first-order filter, and the output error of the first-order filter in a secure environment, respectively.

***Step 1*:** Consider the coordinate transformation $\psi_1^b = \diamondsuit_1(z_1^b, t) := ln\left(\frac{\epsilon_1^b}{1-\epsilon_1^b}\right)$, where $\epsilon_1^b = \frac{z_1^b - \underline{\eta}(t)}{\triangle \eta}$ with $\triangle \eta = \bar{\eta}(t) - \underline{\eta}(t)$. Then, choose the Lyapunov function as

$$V_1^b = \frac{1}{2}(\psi_1^b)^2. \tag{28}$$

By taking the derivative of V_1^b, we can obtain

$$\begin{aligned}\dot{V}_1^b = {} & \psi_1^b \dot{\psi}_1^b \\ = {} & \psi_1^b o^b \Bigg(z_2^b + e_2 + \alpha_1^b + \nu_2^b + f_1(y) - f_1(\hat{y}) + f_1(\hat{y}) \\ & + d_1 - \hat{d}_1 + \hat{d}_1 - \dot{y}_d + \frac{\varpi^b}{o^b} \Bigg), \end{aligned} \tag{29}$$

where $o^b = \frac{1}{\epsilon_1^b(1-\epsilon_1^b)}$, $\varpi^b = \frac{o^b}{\triangle \eta}(\underline{\eta}\dot{\bar{\eta}} - \dot{\underline{\eta}}\bar{\eta} - z_1^b \triangle \dot{\eta})$, $\dot{\bar{\eta}} = (\text{sign}(z_1^b(0)) + \bar{\gamma})\dot{\rho}$ and $\dot{\underline{\eta}} = (\text{sign}(z_1^b(0)) - \underline{\gamma})\dot{\rho}$.

By using Young's inequality, we can get

$$\begin{aligned}\dot{V}_1^b = {} & 3(\psi_1^b)^2(o^b)^2 + \frac{1}{2}(z_2^b)^2 + \frac{1}{2}(\nu_2^b)^2 + \frac{1}{2}\tilde{\Phi}_{1b}^2 + \Lambda_1^b ||e||^2 \\ & + \psi_1^b o^b \left(\alpha_1^b + f_1(\hat{y}) + \hat{d}_1 - \dot{y}_d + \frac{\varpi^b}{o^b} \right), \end{aligned} \tag{30}$$

where $\Lambda_1^b = \frac{1}{2} + \frac{1}{2} l_{1b}^2 + \frac{1}{2}\omega_{1b}^2$.

Then, the virtual controller α_1^b can be designed as

$$\alpha_1^b = -\frac{c_1^b}{o^b}\psi_1^b - 3\psi_1^b o^b - f_1(\hat{y}) + \dot{y}_d - \hat{d}_1 - \frac{\varpi^b}{o^b}, \tag{31}$$

where c_1^b is the positive parameter of the design.

By substituting (31) into (30), one gets

$$\dot{V}_1^b \le -c_1^b(\psi_1^b)^2 + \frac{1}{2}(z_2^b)^2 + \frac{1}{2}(\nu_2^b)^2 + \frac{1}{2}\tilde{\Phi}_{1b}^2 + \Lambda_1^b ||e||^2. \tag{32}$$

Step k **($2 \le k \le n-1$):** From (11), (12) and (24), we have

$$\dot{z}_k^b = z_{k+1}^b + \nu_{k+1}^b + \alpha_k^b + f_k(\bar{y}) + m_{kb}(y^\ell - \hat{y}) + \hat{d}_k - \dot{\theta}_{kf}^b. \tag{33}$$

Design the Lyapunov function as

$$V_k^b = V_{k-1}^b + \frac{1}{2}(z_k^b)^2 + \frac{1}{2}(\nu_k^b)^2. \tag{34}$$

On the basis of (33) and (34), one gets

$$\begin{aligned}\dot{V}_k^b = {} & \dot{V}_{k-1}^b + z_k^b z_{k+1}^b + z_k^b \upsilon_{k+1}^b + z_k^b(f_k(\bar{y}) - f_k(\hat{y})) \\ & + z_k^b m_{kb}(y^\ell - \hat{y}) + z_k^b(\alpha_k^b + f_k(\hat{y}) + \hat{d}_k - \dot{\theta}_{kf}^b) \\ & + \nu_k^b \dot{\nu}_k^b. \end{aligned} \tag{35}$$

By using Young's inequality, the virtual controller α_k^b can be designed as

$$\alpha_k^b = -c_k^b z_k^b - \frac{7}{2} z_k^b - f_k(\hat{y}) - \hat{d}_k + \dot{\theta}_{kf}^b, \tag{36}$$

where c_k^b is the positive parameter of the design.

By substituting (36) into (35), one gets

$$\begin{aligned}V_k^b \le {} & -c_1^b(\psi_1^b)^2 - \sum_{i=2}^{k} c_i^b (z_i^b)^2 - \sum_{i=2}^{k} r_i^b (\nu_i^b)^2 + \frac{1}{2}(z_{k+1}^b)^2 \\ & + \frac{1}{2}(\nu_{k+1}^b)^2 + \frac{1}{2}\tilde{\Phi}_{1b}^2 + \Lambda_k^b ||e||^2 + \frac{1}{2}\sum_{i=1}^{k-1}(\dot{\alpha}_i^b)^2 + \Delta_k^b, \end{aligned} \tag{37}$$

where $\frac{1}{\tau_k^b} \ge 1 + r_k^b$, $\Lambda_k^b = \frac{1}{2} + \frac{1}{2}\sum_{i=1}^{k} l_{ib}^2 + \frac{1}{2}\sum_{i=2}^{k} m_{ib}^2$, and $\Delta_k^b = \frac{1}{2}\sum_{i=2}^{k} l_{ib}^2 \mathfrak{m}_1^2 + \frac{1}{2}\sum_{i=2}^{k} m_{ib}^2 \mathfrak{m}_1^2$.

***Step n*:** From (11), (12) and (24), we have

$$\dot{z}_n^b = u^b + f_n(\bar{y}) + m_{nb}(y^\ell - \hat{y}) + \hat{d}_n - \dot{\theta}_{nf}^b. \tag{38}$$

Design the Lyapunov function as

$$V_n^b = V_{n-1}^b + \frac{1}{2}(z_n^b)^2 + \frac{1}{2}(\nu_n^b)^2. \tag{39}$$

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On the basis of (38) and (39), one gets

$$\dot{V}_n^b = \dot{V}_{n-1}^b + z_n^b(f_n(\bar{y}) - f_n(\hat{y})) + z_n^b m_{nb}(y^\ell - \hat{y}) + z_n^b(u^b + f_n(\hat{y}) + \hat{d}_n - \dot{\theta}_{nf}^b) + \nu_n^b \dot{\nu}_n^b. \quad (40)$$

By using Young's inequality, the controller u^b can be designed as

$$u^b = -c_n^b z_n^b - \frac{5}{2} z_n^b - f_n(\hat{y}) - \hat{d}_n + \dot{\theta}_{nf}^b. \quad (41)$$

where c_n^b is the positive parameter of the design.

Considering the setting of the ETM, we redefine the tracking error in (27) as follows:

$$\begin{aligned} \bar{z}_1^b &= \bar{x}_1 - y_d, \\ \bar{z}_k^b &= \hat{x}_k - \bar{\theta}_{kf}^b, k = 2, \cdots, n \end{aligned} \quad (42)$$

where $\bar{\theta}_{kf}^b$ can be acquired through the designed ETM, that is, $\bar{\theta}_{kf}^b(t) = \theta_{kf}^b(t_{k,s}), \forall t \in [t_{k,s}, t_{k,s+1}), k = 2, \cdots, n$ with

$$t_{k,s+1} = \inf\left\{ t > t_{k,s}, \left| \dot{\theta}_{kf}(t) - \dot{\bar{\theta}}_{kf}(t) \right| > \mathfrak{m}_2 \right\}, \quad (43)$$

where $\mathfrak{m}_2$ is the designed triggering threshold that satisfies $\mathfrak{m}_2 > 0$ and $t_{k,0}$ denotes the first triggering moment at which condition (43) is satisfied.

Then, the event-triggered controllers can be designed to:

$$\bar{\alpha}_1^b = -\frac{c_1^b}{\bar{o}^b}\bar{\psi}_1^b - 3\bar{\psi}_1^b \bar{o}^b - f_1(\hat{y}) + \dot{y}_d - \hat{d}_1 - \frac{\bar{\varpi}^b}{\bar{o}^b}, \quad (44)$$

$$\bar{\alpha}_k^b = -c_k^b \bar{z}_k^b - \frac{7}{2}\bar{z}_k^b - f_k(\hat{y}) - \hat{d}_k + \frac{\bar{\alpha}_{k-1}^b - \bar{\theta}_{kf}^b}{\tau_k^b}, \quad (45)$$

$$u^b = -c_n^b \bar{z}_n^b - \frac{5}{2}\bar{z}_n^b - f_n(\hat{y}) - \hat{d}_n + \frac{\bar{\alpha}_{n-1}^b - \bar{\theta}_{nf}^b}{\tau_n^b}, \quad (46)$$

where $k = 2, \cdots, n$, $\bar{o}^b = \frac{1}{\bar{\epsilon}_1^b(1-\bar{\epsilon}_1^b)}$ with $\bar{\epsilon}_1^b = \frac{\bar{z}_1^b - \eta(t)}{\triangle\eta}$, $\bar{\varpi}^b = \frac{\bar{o}^b}{\triangle\eta}(\underline{\eta}\dot{\bar{\eta}} - \dot{\underline{\eta}}\bar{\eta} - \bar{z}_1^b \triangle\dot{\eta})$.

Remark 4: When the system is subjected to DoS attacks, due to the setting of an output ETM, only the intermittent signal $\bar{y}$ can be used for the actual controller design. Based on conventional backstepping design, the virtual controller $\bar{\alpha}_k^b$ in (45) should be $\bar{\alpha}_k^b = -c_k^b \bar{z}_k^b - \frac{7}{2}\bar{z}_k^b - f_k(\hat{y}) - \hat{d}_k + \dot{\bar{\theta}}_{kf}^b$, but $\dot{\bar{\theta}}_{kf}^b$ is not differentiable. To solve this problem, we design the virtual controller $\bar{\alpha}_k^b$ as $\bar{\alpha}_k^b = -c_k^b \bar{z}_k^b - \frac{7}{2}\bar{z}_k^b - f_k(\hat{y}) - \hat{d}_k + \frac{\bar{\alpha}_{k-1}^b - \bar{\theta}_{kf}^b}{\tau_k^b}$, where $\bar{\alpha}_{k-1}^b$ and $\bar{\theta}_{kf}^b$ are the trigger signals for α_{k-1}^b and θ_{kf}^b, respectively.

Remark 5: Under DoS attacks, MTBFs can still effectively improve the transient performance of the system by constraining the tracking error within prescribed boundaries. However, the coupling effect between MTBFs and the ETM complicates the analysis of triggering errors. Specifically,the error of $\alpha_{k-1}^b(t) - \bar{\alpha}_{k-1}^b(t)$ includes $|\psi_1^b - \bar{\psi}_1^b|$, but the boundary of $|\psi_1^b - \bar{\psi}_1^b|$ is difficult to obtain. To address this issue, we have designed an ETM related to $\dot{\theta}_{kf}^b(t) - \dot{\bar{\theta}}_{kf}^b(t)$ for the output $\theta_{kf}^b(t)$ of the first-order filter in (43).

In order to demonstrate that the designed replacement scheme ensures consensus stability while saving communication resources, the following lemma needs to be proposed.

Lemma 1: The following outlines the upper bounds for triggering errors:

$$|z_1^b - \bar{z}_1^b| \le \mathfrak{m}_1, \quad (47)$$

$$|z_k^b - \bar{z}_k^b| = |\theta_{kf} - \bar{\theta}_{kf}| \le \mathfrak{m}_a, \quad (48)$$

$$\left| \frac{\alpha_{k-1}^b - \theta_{kf}^b}{\tau_k^b} - \frac{\bar{\alpha}_{k-1}^b - \bar{\theta}_{kf}^b}{\tau_k^b} \right| \le \frac{\mathfrak{m}_b + \mathfrak{m}_a}{\tau_k^b}, \quad (49)$$

for $k = 2, \cdots, n$, where $\mathfrak{m}_a$ and $\mathfrak{m}_b$ are positive constants associated with the trigger thresholds $\mathfrak{m}_1$ and $\mathfrak{m}_2$.

Proof: Through (27) and (42), we can obtain

$$|z_1^b - \bar{z}_1^b| \le = |x_1 - \bar{x}_1| \le \mathfrak{m}_1. \quad (50)$$

According to (43), it can be inferred that

$$|\theta_{kf} - \bar{\theta}_{kf}| \le \mathfrak{m}_a, \quad (51)$$

with $t \in [t_{k,s}, t_{k,s+1})$ and $\mathfrak{m}_a$ is positive constant, so we can deduce that

$$|z_k^b - \bar{z}_k^b| = |\theta_{kf} - \bar{\theta}_{kf}| \le \mathfrak{m}_a. \quad (52)$$

According to the ETM (43) and (26), one gets

$$|\alpha_{k-1}^b - \bar{\alpha}_{k-1}^b| \le \mathfrak{m}_b, \quad (53)$$

where $\mathfrak{m}_b$ is positive constant associated with $\mathfrak{m}_a$ and the trigger threshold $\mathfrak{m}_2$.

Based on the above analysis, we can deduce that

$$\begin{aligned} &\left| \frac{\alpha_{k-1}^b - \theta_{kf}^b}{\tau_k^b} - \frac{\bar{\alpha}_{k-1}^b - \bar{\theta}_{kf}^b}{\tau_k^b} \right| \\ &\le \frac{|\alpha_{k-1}^b - \bar{\alpha}_{k-1}^b|}{\tau_k^b} + \frac{|\theta_{kf}^b - \bar{\theta}_{kf}^b|}{\tau_k^b} \\ &\le \frac{\mathfrak{m}_b + \mathfrak{m}_a}{\tau_k^b}. \end{aligned} \quad (54)$$

This completes the demonstration of ***Lemma 1***. ■

From (40) and (46), we can get

$$\begin{aligned} \dot{V}_n^b \le& -c_1^b(\psi_1^b)^2 - \sum_{i=2}^n c_i^b(z_i^b)^2 - \sum_{i=2}^n r_i^b(\nu_i^b)^2 \\ &+ \frac{1}{2}\tilde{\Phi}_{1b}^2 + \Lambda_n^b ||e||^2 \\ &+ \frac{1}{2}\sum_{i=1}^{n-1}(\dot{\alpha}_i^b)^2 + \Delta_{n-1}^b + |z_n^b| \Bigg(\left(c_n^b + \frac{5}{2}\right)(z_n^b - \bar{z}_n^b) \\ &+ \frac{\alpha_{n-1}^b - \theta_{nf}^b}{\tau_n^b} - \frac{\bar{\alpha}_{n-1}^b - \bar{\theta}_{nf}^b}{\tau_n^b} \Bigg) + \frac{1}{2} l_{nb}^2 \mathfrak{m}_1^2 + \frac{1}{2} m_{nb}^2 \mathfrak{m}_1^2, \end{aligned} \quad (55)$$

where $\frac{1}{\tau_n^b} \ge 1 + r_n^b$ and $\Lambda_n^b = \frac{1}{2} + \frac{1}{2}\sum_{i=1}^n l_{ib}^2 + \frac{1}{2}\sum_{i=2}^n m_{ib}^2$.

According to ***Lemma 1*** and (55), we can derive

$$\begin{aligned} \dot{V}_n^b \le& -c_1^b(\psi_1^b)^2 - \sum_{i=2}^n c_i^b(z_i^b)^2 - \sum_{i=2}^n r_i^b(\nu_i^b)^2 + \Lambda_n^b ||e||^2 \\ &+ \frac{1}{2}\sum_{i=1}^{n-1}(\dot{\alpha}_i^b)^2 + \Delta_{n-1}^b + |z_n^b|\varepsilon + \frac{1}{2} l_{nb}^2 \mathfrak{m}_1^2 + \frac{1}{2} m_{nb}^2 \mathfrak{m}_1^2, \end{aligned} \quad (56)$$

where $\varepsilon = \left(c_n^b + \frac{5}{2}\right)\mathfrak{m}_a^2 + \frac{\mathfrak{m}_b + \mathfrak{m}_a}{\tau_n^b}$ is a positive constant.

By using Young's inequality, we can obtain

$$|z_n^b|\varepsilon \le \frac{1}{2}(z_n^b)^2 + \frac{1}{2}\varepsilon^2. \quad (57)$$

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Based on the above inequality (57), we can obtain

$$\dot{V}_n^b \le -c_1^b(\psi_1^b)^2 - \sum_{i=2}^{n}\mathfrak{a}_i^b(z_i^b)^2 - \sum_{i=2}^{n} r_i^b(\nu_i^b)^2 + \frac{1}{2}\tilde{\Phi}_{1b}^2 + \Lambda_n^b||e||^2 + \frac{1}{2}\sum_{i=1}^{n-1}(\dot{\alpha}_i^b)^2 + \Delta_n^b, \tag{58}$$

where $\mathfrak{a}_k^b = c_k^b (k = 2, \cdots, n-1)$, $\mathfrak{a}_n^b = c_n^b - \frac{1}{2}$ and $\Delta_n^b = \frac{1}{2}\sum_{i=2}^{n} l_{ib}^2\mathfrak{m}_1^2 + \frac{1}{2}\sum_{i=2}^{n} m_{ib}^2\mathfrak{m}_1^2 + \frac{1}{2}\varepsilon^2$.

At time $t \in [T_j^b, T_{j+1}^a)$, the overall Lyapunov function can be chosen as

$$V^b = V_0^b + V_n^b = e^T P_b e + \frac{1}{2}\sum_{i=1}^{n}\tilde{\Phi}_{ib}^2 + \frac{1}{2}(\psi_1^b)^2 + \frac{1}{2}\sum_{i=2}^{n}(z_i^b)^2 + \frac{1}{2}\sum_{i=2}^{n}(\nu_i^b)^2. \tag{59}$$

Based on (23) and (58), we can obtain

$$\begin{aligned}\dot{V}^b \le & e^T\left(A_b^T P_b + P_b A_b + P_b\mathcal{L}_b\mathcal{L}_b^T P_b + P_b\mathcal{M}_b\mathcal{M}_b^T P_b + 2P_b^T P_b + \frac{3}{2}\sum_{i=1}^{n}\omega_{ib}^2 + \frac{1}{2}\sum_{i=1}^{n}\omega_{ib}^4 + \Lambda_n^b\right)e - \sum_{i=1}^{n}(q_{ib}-3)\tilde{\Phi}_{ib}^2 \\ & - c_1^b(\psi_1^b)^2 - \sum_{i=2}^{n}\mathfrak{a}_i^b(z_i^b)^2 - \sum_{i=2}^{n} r_i^b(\nu_i^b)^2 + \frac{1}{2}\sum_{i=1}^{n-1}(\dot{\alpha}_i^b)^2 + \Delta_n^b + \mathfrak{V}_b,\end{aligned} \tag{60}$$

where $q_{1b} = \omega_{1b} - \frac{1}{2}$, $q_{kb} = \omega_{kb} (k = 2, \cdots, n)$.

For any $\mathbb{k}_1 > 0$, the set $\Omega_b := \left\{2e^T P_b e + (\psi_1^b)^2 + \sum_{i=2}^{n}\tilde{\Phi}_{ib}^2 + \sum_{i=1}^{n}(z_i^b)^2 + \sum_{i=2}^{n}(\nu_i^b)^2 \le \mathbb{k}_1\right\}$ is a compact one. Thus $|\dot{\alpha}_i^b| \le \mathcal{D}_i^b$ on Ω_b, $i = 1, \cdots, n-1$, From this, we can obtain

$$\begin{aligned}\dot{V}^b \le & e^T\left(A_b^T P_b + P_b A_b + P_b\mathcal{L}_b\mathcal{L}_b^T P_b + P_b\mathcal{M}_b\mathcal{M}_b^T P_b + 2P_b^T P_b + \frac{3}{2}\sum_{i=1}^{n}\omega_{ib}^2 + \frac{1}{2}\sum_{i=1}^{n}\omega_{ib}^4 + \Lambda_n^b\right)e - \sum_{i=1}^{n}(q_{ib}-3)\tilde{\Phi}_{ib}^2 \\ & - c_1^b(\psi_1^b)^2 - \sum_{i=2}^{n}\mathfrak{a}_i^b(z_i^b)^2 - \sum_{i=2}^{n} r_i^b(\nu_i^b)^2 + \Delta_b,\end{aligned} \tag{61}$$

where $\Delta_b = \frac{1}{2}\sum_{i=1}^{n-1}\mathcal{D}_i^b + \Delta_n^b + \mathfrak{V}_b$.

***Case 2*:** When $t \in [T_j^b, T_{j+1}^a)$, the system is under the malicious Dos attack, and at this point, the tracking error (24) becomes

$$z_1^a = \hat{x}_1 - y_d, z_k^a = \hat{x}_k - \theta_{kf}^a, \nu_k^a = \theta_{kf}^a - \alpha_{k-1}^a, \tag{62}$$

where $k = 2, \cdots, n$, $z_k^a (k = 1, \cdots, n)$, α_{k-1}^a, θ_{kf}^a and ν_k^a represent the errors of the system, the virtual controller, the output of the first-order filter, and the output error of the first-order filter under DoS attacks, respectively.

***Step 1*:** Consider the coordinate transformation $\psi_1^a = \Diamond_1(z_1^a, t) := ln\left(\frac{\epsilon_1^a}{1-\epsilon_1^a}\right)$, where $\epsilon_1^a = \frac{z_1^a - \eta(t)}{\triangle\eta}$ with $\triangle\eta = \bar{\eta}(t) - \underline{\eta}(t)$. Then, choose the Lyapunov function as

$$V_1^a = \frac{1}{2}(\psi_1^a)^2. \tag{63}$$

By taking the derivative of V_1^a, we can obtain

$$\dot{V}_1^a = \psi_1^a o^a\left(z_2^a + m_{1a}e_a + \alpha_1^a + \nu_2^a + f_1(\hat{y}) + \hat{d}_1 - \dot{y}_d + \frac{\varpi^a}{o^a}\right), \tag{64}$$

where $o^a = \frac{1}{\epsilon_1^a(1-\epsilon_1^a)}$, $\varpi^a = \frac{o^a}{\triangle\eta}(\underline{\eta}\dot{\bar{\eta}} - \dot{\underline{\eta}}\bar{\eta} - z_1^a\triangle\dot{\eta})$, $\dot{\bar{\eta}} = (\text{sign}(z_1^a(0)) + \bar{\gamma})\dot{\rho}$ and $\dot{\underline{\eta}} = (\text{sign}(z_1^a(0)) - \underline{\gamma})\dot{\rho}$.

By using Young's inequality, we can get

$$\dot{V}_1^a = \frac{3}{2}(\psi_1^a)^2(o^a)^2 + \frac{1}{2}(z_2^a)^2 + \frac{1}{2}(\nu_2^a)^2 + \frac{1}{2}m_{1a}^2||e||^2 + \psi_1^a o^a\left(\alpha_1^a + f_1(\hat{y}) + \hat{d}_1 - \dot{y}_d + \frac{\varpi^a}{o^a}\right). \tag{65}$$

The, the virtual controller α_1^a can be designed as

$$\alpha_1^a = -\frac{c_1^a}{o^a}\psi_1^a - \frac{3}{2}\psi_1^a o^a - f_1(\hat{y}) + \dot{y}_d - \hat{d}_1 - \frac{\varpi^a}{o^a}, \tag{66}$$

where c_1^a is the positive parameter of the design.

By substituting (66) into (64), one gets

$$\dot{V}_1^a \le -c_1^a(\psi_1^a)^2 + \frac{1}{2}(z_2^a)^2 + \frac{1}{2}(\nu_2^a)^2 + \frac{1}{2}m_{1a}^2||e||^2. \tag{67}$$

Step k $(2 \le k \le n-1)$**:** Design the Lyapunov function as

$$V_k^a = V_{k-1}^a + \frac{1}{2}(z_k^a)^2 + \frac{1}{2}(\nu_k^a)^2. \tag{68}$$

On the basis of (11), (12), (24) and (68), one gets

$$\dot{V}_k^a = \dot{V}_{k-1}^a + z_k^a z_{k+1}^a + z_k^a v_{k+1}^a + z_k^a m_{ka}e_a + z_k^a(\alpha_k^a + f_k(\hat{y}) + \hat{d}_k - \dot{\theta}_{kf}^b) + \nu_k^a\dot{\nu}_k^a. \tag{69}$$

Based on (69) and Young's inequality, the virtual controller α_k^a can be designed as

$$\alpha_k^a = -c_k^a z_k^a - 2z_k^a - f_k(\hat{y}) - \hat{d}_k + \dot{\theta}_{kf}^a, \tag{70}$$

where c_k^a is the positive parameter of the design.

By substituting (70) into (69), one gets

$$V_k^a \le -c_1^a(\psi_1^a)^2 - \sum_{i=2}^{k} c_i^a(z_i^a)^2 - \sum_{i=2}^{k} r_i^a(\nu_i^a)^2 + \frac{1}{2}(z_{k+1}^a)^2 + \frac{1}{2}(\nu_{k+1}^a)^2 + \frac{1}{2}\sum_{i=1}^{k} m_{ia}^2||e||^2 + \frac{1}{2}\sum_{i=1}^{k-1}(\dot{\alpha}_i^a)^2, \tag{71}$$

where $\frac{1}{\tau_k^a} \ge 1 + r_k^a$.

***Step n*:** Design the Lyapunov function as

$$V_n^a = V_{n-1}^a + \frac{1}{2}(z_n^a)^2 + \frac{1}{2}(\nu_n^a)^2. \tag{72}$$

On the basis of (11), (12), (24) and (72), one gets

$$\dot{V}_n^a = \dot{V}_{n-1}^a + z_n^a m_{na}e_a + z_n^a(u^a + f_n(\hat{y}) + \hat{d}_n - \dot{\theta}_{nf}^a) + \nu_n^a\dot{\nu}_n^a. \tag{73}$$

Based on (73) and Young's inequality, the actual controller u^a can be designed as

$$u^a = -c_n^a z_n^a - z_n^a - f_n(\hat{y}) - \hat{d}_n + \dot{\theta}_{nf}^a, \tag{74}$$

where c_n^a is the positive parameter of the design.

By substituting (74) into (73), one gets

$$V_n^a \le -c_1^a(\psi_1^a)^2 - \sum_{i=2}^{n}\mathfrak{a}_i^a(z_i^a)^2 - \sum_{i=2}^{n} r_i^a(\nu_i^a)^2$$

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$$+\frac{1}{2}\sum_{i=1}^{k} m_{ia}^2||e||^2+\frac{1}{2}\sum_{i=1}^{n-1}(\dot{\alpha}_i^a)^2, \tag{75}$$

where $\frac{1}{\tau_n^a}\geq 1+r_n^a$, $\mathfrak{a}_k^a=c_k^a(k=2,\cdots,n)$.

At time $t\in[T_j^b,T_{j+1}^a)$, the overall Lyapunov function can be chosen as

$$V^a=e^TP_ae+\frac{1}{2}\sum_{i=1}^{n}\tilde{\Phi}_{ia}^2+\frac{1}{2}(\psi_1^a)^2+\frac{1}{2}\sum_{i=2}^{n}(z_i^a)^2+\frac{1}{2}\sum_{i=2}^{n}(\nu_i^a)^2. \tag{76}$$

Based on (20) and (75), we can obtain

$$\begin{aligned}\dot{V}^a\leq\ & e^T\Bigg(A_a^TP_a+P_aA_a+\frac{1}{k}+k\left\|P_a\mathcal{L}_a\right\|^2+\frac{1}{c}+c\left\|P_a\mathcal{M}_a\right\|^2\\&+2P_a^TP_a+\frac{3}{2}\sum_{i=1}^{n}\omega_{ia}^2+\frac{1}{2}\sum_{i=1}^{n}\omega_{ia}^2l_{ia}^2+\frac{1}{2}\sum_{i=1}^{n}\omega_{ia}^4\\&+\frac{1}{2}\sum_{i=1}^{k}m_{ia}^2\Bigg)e-\sum_{i=1}^{n}(\omega_{ia}-3)\tilde{\Phi}_{ia}^2-\sum_{i=2}^{n}\mathfrak{a}_i^a(z_i^a)^2\\&-c_1^a(\psi_1^a)^2-\sum_{i=2}^{n}r_i^a(\nu_i^a)^2+\frac{1}{2}\sum_{i=1}^{n-1}(\dot{\alpha}_i^a)^2+\mathfrak{V}_a.\end{aligned} \tag{77}$$

For any $\mathbb{k}_2>0$, the set $\Omega_a:=\left\{2e^TP_ae+(\psi_1^a)^2+\sum_{i=2}^{n}\tilde{\Phi}_{ia}^2+\sum_{i=1}^{n}(z_i^a)^2+\sum_{i=2}^{n}(\nu_i^a)^2\leq\mathbb{k}_2\right\}$ is a compact one. Thus $|\dot{\alpha}_i^a|\leq\mathcal{D}_i^a$ on Ω_a, $i=1,\cdots,n-1$. From this, we can obtain

$$\begin{aligned}\dot{V}^a\leq\ & e^T\Bigg(A_a^TP_a+P_aA_a+\frac{1}{k}+k\left\|P_a\mathcal{L}_a\right\|^2+\frac{1}{c}+c\left\|P_a\mathcal{M}_a\right\|^2\\&+2P_a^TP_a+\frac{3}{2}\sum_{i=1}^{n}\omega_{ia}^2+\frac{1}{2}\sum_{i=1}^{n}\omega_{ia}^2l_{ia}^2-\sum_{i=1}^{n}(\omega_{ia}-3)\tilde{\Phi}_{ia}^2\\&+\frac{1}{2}\sum_{i=1}^{n}\omega_{ia}^4+\frac{1}{2}\sum_{i=1}^{k}m_{ia}^2\Bigg)e-c_1^a(\psi_1^a)^2-\sum_{i=2}^{n}\mathfrak{a}_i^a(z_i^a)^2\\&-\sum_{i=2}^{n}r_i^a(\nu_i^a)^2+\Delta_a,\end{aligned} \tag{78}$$

where $\Delta_a=\frac{1}{2}\sum_{i=1}^{n-1}\mathcal{D}_i^2+\mathfrak{V}_a$.

In addition, the estimated design parameter vectors $\mathcal{M}_b$ and $\mathcal{M}_a$ are selected to satisfy the following inequality:

$$\begin{aligned}&A_b^TP_b+P_bA_b+P_b\mathcal{L}_b\mathcal{L}_b^TP_b+P_b\mathcal{M}_b\mathcal{M}_b^TP_b+2P_b^TP_b\\&\quad+\frac{3}{2}\sum_{i=1}^{n}\omega_{ib}^2+\frac{1}{2}\sum_{i=1}^{n}\omega_{ib}^4+\Lambda_n^b\leq-bP_b\end{aligned} \tag{79}$$

and

$$\begin{aligned}&A_a^TP_a+P_aA_a+\frac{1}{k}+k\left\|P_a\mathcal{L}_a\right\|^2+\frac{1}{c}+c\left\|P_a\mathcal{M}_a\right\|^2+2P_a^TP_a\\&\quad+\frac{3}{2}\sum_{i=1}^{n}\omega_{ia}^2+\frac{1}{2}\sum_{i=1}^{n}\omega_{ia}^2l_{ia}^2+\frac{1}{2}\sum_{i=1}^{n}\omega_{ia}^4+\frac{1}{2}\sum_{i=1}^{k}m_{ia}^2\\&\leq aP_a.\end{aligned} \tag{80}$$

Therefore, (61) and (78) can be expressed as

$$\dot{V}^b\leq-be^TP_be-\sum_{i=1}^{n}(q_{ib}-3)\tilde{\Phi}_{ib}^2-c_1^b(\psi_1^b)^2-\sum_{i=2}^{n}\mathfrak{a}_i^b(z_i^b)^2-\sum_{i=2}^{n}r_i^b(\nu_i^b)^2+\Delta_b \tag{81}$$

and

$$\dot{V}^a\leq ae^TP_ae-\sum_{i=1}^{n}(\omega_{ia}-3)\tilde{\Phi}_{ia}^2-c_1^a(\psi_1^a)^2-\sum_{i=2}^{n}\mathfrak{a}_i^a(z_i^a)^2-\sum_{i=2}^{n}r_i^a(\nu_i^a)^2+\Delta_a. \tag{82}$$

At the same time, for $i=2,\cdots,n$ and $j=1,\cdots,n$, choose the design parameters ω_{ja}, ω_{jb}, c_1^a, c_1^b, $\mathfrak{a}_i^a$, $\mathfrak{a}_i^b$, r_i^a and r_i^b as

$$\begin{cases}-2\sum_{i=1}^{n}(\omega_{ia}-3)<a,\\-2c_1^a<a,\\-2\mathfrak{a}_i^a<a,\\-2r_i^a<a,\end{cases} \tag{83}$$

and

$$\begin{cases}-2\sum_{i=1}^{n}(q_{ib}-3)<-b,\\-2c_1^b<-b,\\-2\mathfrak{a}_i^b<-b,\\-2r_i^b<-b.\end{cases} \tag{84}$$

Based on (81)-(84), we can obtain

$$\begin{aligned}\dot{V}^a&\leq aV^a+\Delta_a,\\\dot{V}^b&\leq-bV^b+\Delta_b.\end{aligned} \tag{85}$$

As a result,the derivative of the overall Lyapunov function can be formulated as

$$\dot{V}=\begin{cases}\dot{V}^a, & t\in[T_j^a,T_j^b)\\\dot{V}^b, & t\in[T_j^b,T_{j+1}^a)\end{cases} \tag{86}$$

C. System Stability

Theorem 1: The considered system (1) with output event-triggered under DoS attacks and satisfies ***Assumption 1-5***, including the virtual controllers (44), (45), (66), (70), the actual controllers (46), (74) and the observer (11). If positive definite matrices P_a and P_b, vectors $\mathcal{M}_a$ and $\mathcal{M}_b$, and constant $\eth>0$ exist and satisfy the following inequality:

$$\begin{aligned}&P_a<\wp P_b,P_b<\wp P_a,\\&\frac{b-\eth}{b+a}>\frac{1}{\mathcal{T}_1},\eth>\frac{\ln\wp}{\epsilon_D}.\end{aligned} \tag{87}$$

As a result, the designed resilient intermittent tracking control scheme guarantees that all signals in the closed-loop system remain bounded, and the tracking error converges to a prescribed neighborhood.

Proof: Based on (85)-(87), we are able to acquire

$$\dot{V}\leq\hbar(t)V+\Delta, \tag{88}$$

where $\hbar(t)=a$ when $t\in[T_j^a,T_j^b)$, and $\hbar(t)=-b$ when $t\in[T_j^b,T_{j+1}^a)$, $\Delta=\Delta_a+\Delta_b$. Then, in the case of $t\in[T_j^a,T_j^b)$, we can derive

$$V(t)\leq\delta e^{\hbar_j\left(t-T_j^a\right)}V(T_j^{a-})+\Delta\int_{T_j^a}^{t}e^{\hbar_j(t-\epsilon)}d\epsilon, \tag{89}$$

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where $\hbar_j$ is a constant that depends on a or b.

A recursive method can be applied during the jth DoS attack interval to obtain

$$V(t) \leq \delta^{\sigma(0,t)} e^{\hbar_j (t-T_j^a)+\cdots+\hbar_0 (T_1^a-0)} V(0) + \Delta \int_0^{T_1^a} \delta^{\sigma(T_1^a,t)} e^{\hbar_j (t-T_j^a)+\cdots+\hbar_0 (T_1^a-\epsilon)} d\epsilon + \cdots + \Delta \int_{T_j^a}^{t} e^{\hbar_j (t-\epsilon)} d\epsilon. \quad (90)$$

Based on ***Assumption 4***, for the time interval $[\epsilon, t]$, we can conclude that $|\beth_a(\epsilon,t)| \leq (t-\epsilon)/\mathcal{T}_1$ and $|\beth_a(\epsilon,t)| \geq (t-\epsilon)/(1-(1/\mathcal{T}_1))$. Furthermore, we can obtain the following inequality

$$V(t) \leq e^{-[b|\beth_b(0,t)|-a|\beth_a(0,t)|-\sigma(0,t)\ln\delta]} V(0) + \Delta \int_0^t e^{-[b|\beth_b(\epsilon,t)|-a|\beth_a(\epsilon,t)|-\sigma(\epsilon,t)\ln\delta]} d\epsilon. \quad (91)$$

Based on ***Assumption 5*** and (87), there exists a constant ζ that makes $b|\beth_b(\epsilon,t)| - a|\beth_a(\epsilon,t)| > \eth(t-\epsilon) > (\sigma(\epsilon,t)-\theta)\ln\delta + \zeta(t-\epsilon)$. Then, we have

$$V(t) \leq \delta^{\theta}\left[e^{-\zeta t}V(0) + \Delta\int_0^t e^{-\zeta(t-\epsilon)}d\epsilon\right] \leq \delta^{\theta}\left[e^{-\zeta t}V(0) + \frac{\Delta}{\zeta}\right]. \quad (92)$$

Also, we can deduce

$$\dot{V} \leq -\zeta V(t) + \Delta. \quad (93)$$

Consequently, the following inequality holds

$$V(t) \leq V(0) + \frac{\Delta}{\zeta}. \quad (94)$$

Based on the above discussion, it can be concluded that the ETC strategy designed when the system is subjected to DoS can ensure that all signals in the closed-loop system are bounded. Therefore, $\psi_1^a = ln\left(\frac{\epsilon_1^a}{1-\epsilon_1^a}\right)$ is bounded, which means $0 < \epsilon_1^a = \frac{z_1^a - \underline{\eta}(t)}{\bar{\eta}(t)-\underline{\eta}(t)} < 1$, i.e. $\bar{\eta}(t) < z_1^a(t) < \underline{\eta}(t)$. Similarly, $\bar{\eta}(t) < z_1^b(t) < \underline{\eta}(t)$ can also be obtained. This completes the demonstration of ***Theorem 1***. ■

Finally, we verify that Zeno's behavior is eliminated. Define $\Theta_s(t) = y(t) - \bar{y}(t), \forall t \in [t_s, t_{s+1})$ and $\Theta_{i,s}(t) = \dot{\theta}_{if}^b(t) - \dot{\bar{\theta}}_{if}^b(t), \forall t \in [t_{i,s}, t_{i,s+1})$, $i = 1, \cdots, n$, whose derivatives are computed as

$$\frac{d}{dt}|\Theta_s| = \frac{d}{dt}(\Theta_s \times \Theta_s)^{\frac{1}{2}} = \operatorname{sign}(\Theta_s)\dot{\Theta}_s^i \leq |\dot{y}|, \quad (95)$$

$$\frac{d}{dt}|\Theta_{i,s}| = \frac{d}{dt}(\Theta_{i,s} \times \Theta_{i,s})^{\frac{1}{2}} = \operatorname{sign}(\Theta_{i,s})\dot{\Theta}_{i,s} \leq \left|\ddot{\theta}_{if}^b\right|. \quad (96)$$

According to ***Theorem 1***, we can know that x_i, α_i^b, $\dot{\alpha}_i^b$, ν_n^b and θ_{if}^b are all bounded. Therefore, there must be constants $\partial > 0$ and $\lambda_i > 0$ that satisfy $|\dot{y}| \leq \partial$ and $|\ddot{\vartheta}_{if}^b| \leq \lambda_i$. Since $\Theta_s(t_s) = 0$ and $\Theta_{i,s}(t_{i,s}) = 0$, we can obtain $t_{s+1} - t_s \geq \frac{\mathfrak{m}_1}{\partial}, t_{i,s+1} - t_{i,s} \geq \frac{\mathfrak{m}_2}{\lambda_i}$. This implies that Zeno's behavior is avoided.

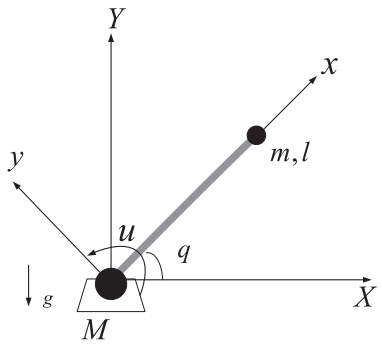


Fig. 3. Framework of single-link robot.

Remark 6: When the system is subjected to malicious DoS attacks, compared with [41], which only considers ETM between the controller and the actuator, this study introduces an ETM in the sensor-to-controller channel, thereby further reducing the communication burden. In contrast to [42], which does not consider external disturbances, the proposed observer is capable of simultaneously estimating the system states and unknown external disturbances. Moreover, by introducing MTBFs to constrain the tracking error, the transient performance of the system is further improved; meanwhile, the dynamic characteristics of a first-order filter are utilized to prove the boundedness of the effects induced by event triggering.

Remark 7: The tracking performance of the system mainly depends on the parameters c_i^b, c_i^a, ω_{ib}, l_{ib}, m_{ib} and $\mathfrak{m}_1$. Appropriately increasing c_i^b, c_i^a and decreasing ω_{ib}, l_{ib}, m_{ib}, $\mathfrak{m}_1$ can reduce the tracking error. However, a smaller $\mathfrak{m}_1$ leads to a higher triggering frequency, thereby weakening the communication-saving effect of the ETM. Therefore, a balance between tracking performance and communication resource consumption should be considered in parameter selection.

Remark 8: In practical applications, implementation issues such as sensor degradation, actuator faults, noise, and real-time limitations may further degrade system performance and affect the reliability of the control system. Therefore, extending the results of this paper to resilient control problems under multiple faults, network attacks, and implementation constraints is a promising direction for future research.

IV. Simulation Studies

This section simulates a single-link robot (see Fig. 3) with output event-triggered under DoS attacks, which can be described as [43]:

Taking into account the external disturbances on a single-link robot [45], the mathematical model is as follows:

$$M\ddot{q} + \frac{1}{2}mgl\sin(q) = u(t) + d(t), y = q, \quad (97)$$

where $M = 0.5$, $g = 9.8$, $m = 1$, $l = 1$, $d(t)$ represents the external disturbance.

In the simulation, the designed desired trajectory $y_d = 1.25\cos(t - 0.6)$, the external disturbance $d(t) = 1.25\cos(0.08t)$. Choose $x_1(0) = 0.8$, $x_2(0) = 0.8$ as the initial states, the triggering thresholds $\mathfrak{m}_1 = 0.54$, $\mathfrak{m}_2 = 0.05$, the parameters $c_1^a = c_1^b = 20$, $c_2^a = c_2^b = 6$,

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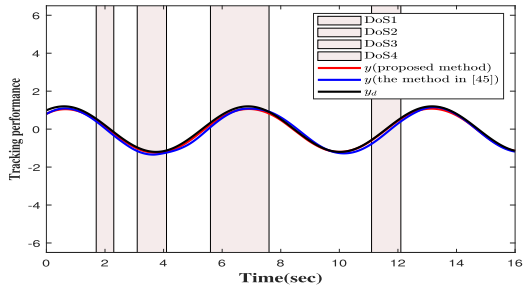


Fig. 4. The trajectories of output signal y and reference signal y_d under different methods.

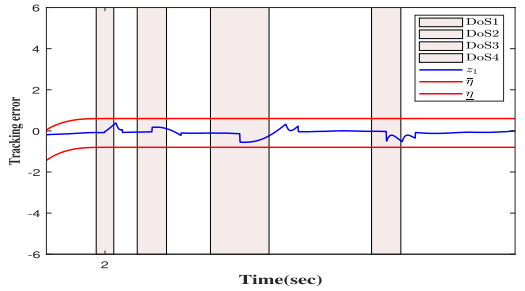


Fig. 5. Output tracking errors.

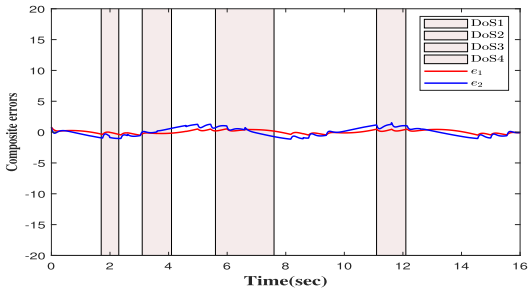


Fig. 6. Observation errors of states x_i for $i = 1, 2$.

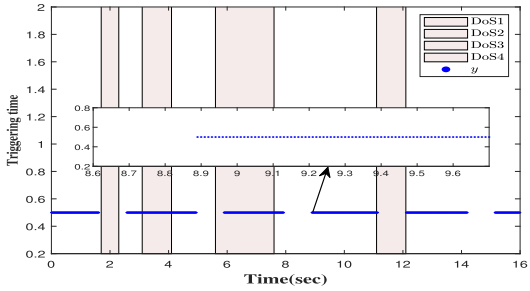


Fig. 7. Triggering instants of y.

TABLE I

COMPARISON OF THE TRIGGER COUNTS OF THE SYSTEM OUTPUT y BETWEEN THE METHOD IN [45] AND THE METHOD PROPOSED IN THIS PAPER

Method / Trigger counts	This paper	The literature [45]
Trigger counts	1263	1600

$l_1^a = l_1^b = 10$, $l_2^a = l_2^b = 23$, $w_{1a} = w_{1b} = 0.95$, $w_{2a} = w_{2b} = 0.13$, $\tau_2^a = \tau_2^b = 104$. Choose the parameters associated with monotonic tubes as $\rho_0 = 1.3$, $\rho_\infty = 10$, $\imath = 4.2$, $T = 3$, $\bar{\gamma} = 0.06$, $\underline{\gamma} = 0.08$.

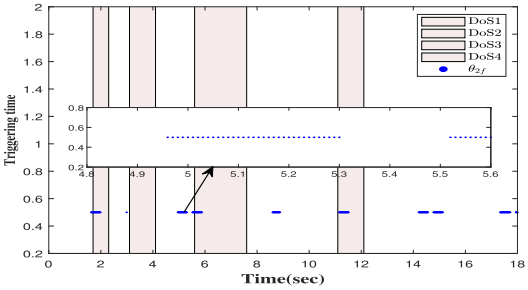


Fig. 8. Triggering instants of θ_{2f}.

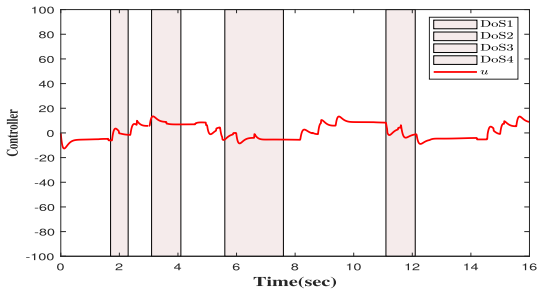


Fig. 9. The trajectory of control input u.

To demonstrate the effectiveness and superiority of the proposed control strategy, we carried out a comparison with the secure control scheme presented in [44]. Simulations are carried out using the same initial conditions and design parameters with a time step of 0.01s. Simulation results are presented in Figs. 4–9, respectively. Fig. 4 represents the tracking performance of the system output y with respect to the desired trajectory y_d under different control methods. It can be observed that the proposed control method achieves better tracking performance than the method in [44] under DoS attacks. Fig. 5 represents the tracking errors of the system. It is worth noting that as the density and duration of DoS attacks increase, the transient performance of the system is slightly degraded, but the tracking error still remains within the prescribed error bound. Fig. 6 displays the composite observer errors. Fig. 7 and Fig. 8 respectively display the triggering instants of y and θ_{2f}. Fig. 9 displays the trajectory of controller u. Furthermore, Table I presents the numbers of triggering events of the system output y under the proposed control method and the method in [44]. It can be seen that the proposed method requires fewer triggering events, thereby reducing communication resource consumption.

V. CONCLUSION

This paper has proposed a dynamic event-triggered prescribed-time adaptive resilient tracking control strategy for a class of nonlinear CPSs subject to malicious DoS attacks. We utilized a set of adaptive high-gain filters established to produce the necessary state estimations to conquer the effects brought about by DoS attacks. In addition, by introducing a dynamic ETM, which in turn reduced the frequency of controller updates, the network communication resources were saved. On the basis of the above, a novel prescribed-time safety controller was presented by integrating a dynamic time-varying constraint function. The above control scheme not only

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guaranteed that the tracking was achieved in the predefined time but also reached the asymptotic convergence. It should be noted that the proposed control framework in this paper permits DoS attacks pattern with arbitrarily varying frequency and duration, so long as they remain bounded. Future research will aim to generalize this approach to nonlinear CPSs operating within more complex networked environments.

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